

Video Compression Homework 1 Report

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Project Link: <https://github.com/RyanCheng98153/video-compression/tree/main/hw1>

Objective

This assignment aims to explore color space transformations using a given image (`lena.png`).

The task includes:

- Converting the image from RGB to:
 - YUV color space
 - YCbCr color space
- Generating grayscale images of the individual channels:
 - $RGB \rightarrow R, G, B$
 - $YUV \rightarrow Y, U, V$
 - $YCbCr \rightarrow Y, Cb, Cr$

All transformations are done manually using the provided formulas without using any built-in color conversion functions.

Input Example: lena.png

- *Input Figure: lena.png*



1. RGB Components

Due to OpenCV's default image reader setting being BGR, a BGR-to-RGB conversion is required for convenience before further processing.

Each color channel (R, G, B) was extracted and saved as a grayscale image.

- *Figure 1: Red Channel*



- *Figure 2: Green Channel*



- *Figure 3: Blue Channel*



2. YUV Transformation

The RGB to YUV transformation uses the following formulas:

$$Y = 0.299 \cdot R + 0.587 \cdot G + 0.114 \cdot B$$

$$U = -0.169 \cdot R - 0.331 \cdot G + 0.5 \cdot B + 128$$

$$V = 0.5 \cdot R - 0.419 \cdot G - 0.081 \cdot B + 128$$

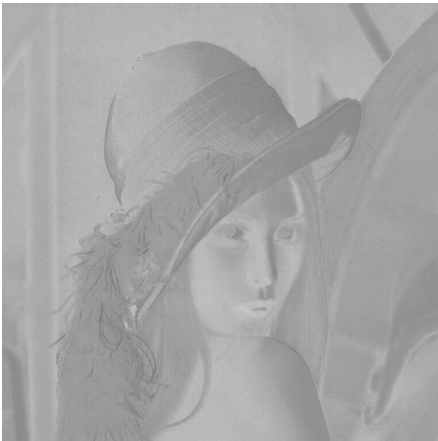
- *Figure 4: Y Channel (YUV)*



- *Figure 5: U Channel (YUV)*



- *Figure 6: V Channel (YUV)*



3. YCbCr Transformation

The RGB to YCbCr transformation uses the following formulas:

$$Y = 0.299 \cdot R + 0.587 \cdot G + 0.114 \cdot B$$

$$Cb = -0.168736 \cdot R - 0.331264 \cdot G + 0.5 \cdot B + 128$$

$$Cr = 0.5 \cdot R - 0.418688 \cdot G - 0.081312 \cdot B + 128$$

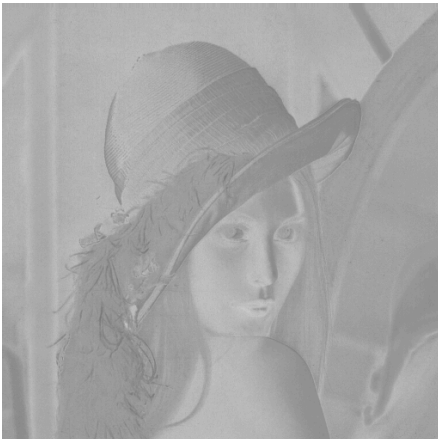
- *Figure 7: Y Channel (YCbCr)*



- *Figure 8: Cb Channel (YCbCr)*



- *Figure 9: Cr Channel (YCbCr)*



4. Implementation Notes

The project dependencies can be managed either through a `requirements.txt` file or by using `uv` (ref: [README.md](#)).

For the implementation, I implemented with "native-python" (readable) and "NumPy" (fast) version:

`./main.py`:

- implemented in native Python with clear and readable logic.

`./numpy_main.py`:

- implemented using NumPy arrays to achieve accelerated computation.

![[NOTE]]

YUV and YCbCr still have a little difference,
can use `np.abs(yuv.astype(int) - ycbcr.astype(int))`
to check the mean and max difference between.

5. Conclusion

This assignment demonstrates manual implementation of **RGB** to **YUV** and **YCbCr** transformations and visualization of each color channel. Understanding the transformation formulas helps reinforce how image data is represented and manipulated in different color spaces.